Samarth Raman Ghanate

Computer Vision & Perception Engineer | Medical Imaging AI | Sensor Fusion
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Skills

Computer Vision & AI: YOLO-World, CLIP, SegFormer, U-Net, PyTorch, TensorFlow, Semantic Segmentation, Open-Vocabulary Detection, Vision-Language Models

Sensor Integration: LiDAR, RGB-D, Depth Cameras, Multi-Sensor Fusion, Point Cloud Processing (Open3D)

Programming: C++17, Python, C#, YAML, Git

Robotics & Middleware: ROS 2 Humble, RViz2, RTAB-Map, Gazebo

Simulation & Digital Twin: Unity3D, Unity-ROS 2 Bridge, Meta Quest 2 / XR

Development Environment: Linux (Ubuntu 22.04 ARM64), Docker, Visual Studio Code

Publications

*M. Kalenberg, M. Heine, **S. R. Ghanate**, C. Hofmann, J. Franke, J. Walter, J. Fuerst**, "Social Force-Based Proxemics for Mutual and Comfortable Navigation of an Intelligent Wheelchair,"** IEEE International Conference on Advanced Robotics and its Social Impact (ARSO 2026), Vienna, Austria, June 2026.*
https://ieeexplore.ieee.org/abstract/document/11536118

Work Experience

Siemens Healthineers — Erlangen, Germany

Research Assistant | Working Student — **02/2025 – Present**

– Built a multi-sensor perception pipeline integrating LiDAR, RGB-D, and camera streams under real-time constraints, eliminating coverage blind spots across 3 sensor modalities and reducing untracked agent incidents in the monitored zone.
– Designed and deployed a Kalman filter-based multi-object tracking system for dynamic agent state estimation, improving trajectory prediction consistency by ~40% over frame-by-frame detection across consecutive observation windows.
– Developed deterministic C++ components for synchronized data acquisition and sensor calibration, cutting pipeline latency by ~30% through multithreaded execution and lock-free queue management.

Hospital Digital Twin & Real-to-Sim Research

– Collaborated with Uniklinikum radiologists and cross-functional engineering teams to translate clinical workflow requirements into simulation constraints, informing sensor placement decisions for robotic applications.
– Integrated high-level robotic decision logic with low-level perception outputs, ensuring coordinated multi-agent behavior under strict timing requirements in a safety-critical environment.
– Developed structured benchmarking and evaluation frameworks to validate perception robustness across sensor configurations, supporting iterative system refinement across real and simulated environments.

Master Thesis: Sensor Selection and Positioning Analysis for Medical Robotics. — **06/2025 – 11/2025**

– Built a high-fidelity Unity-based hospital digital twin to simulate, evaluate, and compare robotic sensor configurations across realistic spatial layouts, reducing physical prototype iteration cycles by enabling virtual pre-validation.
– Implemented synchronized simulation of RGB, depth, and 3D LiDAR sensors in Unity, modeling real-world sensor behaviour including occlusions, noise characteristics, and range limitations across multiple camera placements.
– Integrated Unity with ROS 2 via the Unity-ROS 2 bridge to record time-aligned multimodal sensor data, feeding directly into downstream computer vision and perception pipelines for robotic processing.
– Conducted structured sensor placement studies comparing spatial coverage and detection quality across 6+ layout configurations, producing quantitative trade-off analyses to guide hardware selection decisions.
– Evaluated adaptive perception strategies under varying environmental conditions, informing sensor fusion architecture choices for dynamic, human-occupied clinical spaces.

SAA AB Engineering Pvt Ltd — Bengaluru, India
New Product Development Engineer — **09/2021 – 07/2022**

Research & Project Experience

Semantic Bot: Open Vocabulary Semantic 3D Mapping for Indoor Robots
https://github.com/Samarthraman01/Semanticbot
Independent R&D Project

- Developed an end-to-end ROS 2 Humble perception pipeline integrating YOLO-World and OpenAI CLIP (512-dim embeddings) for open-vocabulary semantic 3D mapping, enabling object detection and classification without retraining on new categories.
- Implemented a C++ ROS 2 fusion node subscribing to depth image and detection topics, applying pinhole projection and SE(3) pose transforms, publishing PointCloud2 on /semantic/pointcloud in real time.
- Built modular Python pipeline (frame_reader, detect_objects, detect_and_lift_3d, clip_embeddings, query_map) producing a semantic map of 688 classified objects queryable by natural language.
- Containerized the full system using Docker (Ubuntu 22.04 ARM64) with structured latency and robustness logging for deployment validation.
- Designed data and software pipeline covering image acquisition, model inference, 3D fusion, and structured output suitable for training, testing, and iterative deployment.

EgoSeg-RT https://github.com/Samarthraman01/EgoSeg-RT
Independent R&D Project

- Conducted a systematic deployment characterization of real-time semantic segmentation on egocentric first-person video from wearable cameras, benchmarking SegFormer-B0 and B2 across PyTorch, ONNX FP32, and INT8 export formats on CPU, MPS, and T4 GPU hardware.
- Quantified a catastrophic domain shift between standard indoor benchmarks and egocentric footage: ADE20K mIoU dropped from 0.350 to IoU = 0.000 on VISOR-HOS (272K annotated egocentric kitchen masks), with failure modes including viewpoint shift, scale shift, and hand blindness.
- Identified SegFormer-B0 as the only model variant achieving real-time throughput (79.4 FPS on T4 GPU, 21.1 FPS on Apple M2 MPS), and showed hardware choice matters more than export format (B2 CPU-to-MPS: 7x speedup vs ONNX: 1.7x speedup).
- Exported a trained B0 checkpoint to ONNX for portable deployment; full reproducible pipeline and benchmark results publicly available on GitHub.

Institute for Factory Automation and Production Systems (FAPS) — Erlangen, Germany
Research Assistant — **07/2024 – 02/2025**

- Developed a Unity-based VR simulator for assistive robotics applications, covering full design, implementation, testing, and iterative refinement cycles.
- Implemented interaction logic, movement systems, and visualization pipelines within Unity for real-time immersive evaluation.
- Delivered a reproducible AR/VR-enabled simulation framework suitable for academic demonstrations and ongoing research development.
- Integrated research into assistive robotics with applications toward medical device navigation and human-robot collaborative safety.

Education

Friedrich Alexander University (FAU) — Erlangen - Nürnberg, Germany
M.Sc. Electromobility Engineering — **10/2022 – 06/2026**

People's Education Society University — Bangaluru, India
B.Tech. Mechanical Engineering — **08/2017 – 08/2021**

Languages

- English – C1 (Professional working proficiency)
- German – B1 (currently improving)